

Execution Plan im Griff:



SQL-Ausführung optimieren ohne Änderung am Code



Über Otto Group one.O

Zusammenschluss zu one.O
2025

Muttergesellschaft
Otto Group

Standorte
Dresden, Hamburg, Altenkunstadt, Madrid,
Málaga, Valencia, Taipeh, Hyderabad

Anzahl Mitarbeitende
rund 1.000

Geschäftsführung
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Zur Person

Peter Ramm



Software Architekt / Teamleiter bei OSP in Dresden

> 30 Jahre Historie in IT-Projekten

Schwerpunkte:

- Entwicklung von OLTP-Systemen auf Basis von Oracle-Datenbanken
- Architektur-Beratung bis Trouble-Shooting
- Performance-Optimierung bestehender Systeme

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Motivation

Warum sollte man sich überhaupt mit dem Thema befassen?

- **Schnelle Reaktion auf Probleme notwendig**
Lösung von inhaltlichen und Laufzeitproblemen kann extrem schnell erforderlich sein
- **Fremdprogramme**
Externe Software ohne eigenen Einfluss auf den ausgeführten Code
- **Problemlösung in Applikation dauert zu lange**
Zeitlicher Aufwand für Analyse der Ursachen und Fixen der Probleme im Code ist nicht akzeptabel

Varianten der
Oracle-DB für
Beeinflussung
der SQL-
Ausführung
ohne
Änderungen am
Code

- **Optimizer Environment**
Ändern per ALTER SESSION bzw. ALTER SYSTEM
z.B. OPTIMIZER_INDEX_COST_ADJ
- **Objekt-Statistiken**
Analyse von Tabellen und Indizes per DBMS_STATS, Histogramme, Extended Statistics
- **Stored Outlines**
Fixierter Execution Plan, Deprecated seit 11g
- **SQL Plan Baselines**
Präferierter Execution Plan über seinen Hash-Wert
- **SQL Profiles**
Injektion von Optimizer-Hints in konkretes SQL Statement
- **SQL Patches**
Analog SQL Profile, mit reduzierter Funktion und Lizenzbedarf
- **SQL Translations**
Transformation von SQL-Text

Genutztes Analyse-Tool: Panorama for Oracle Databases

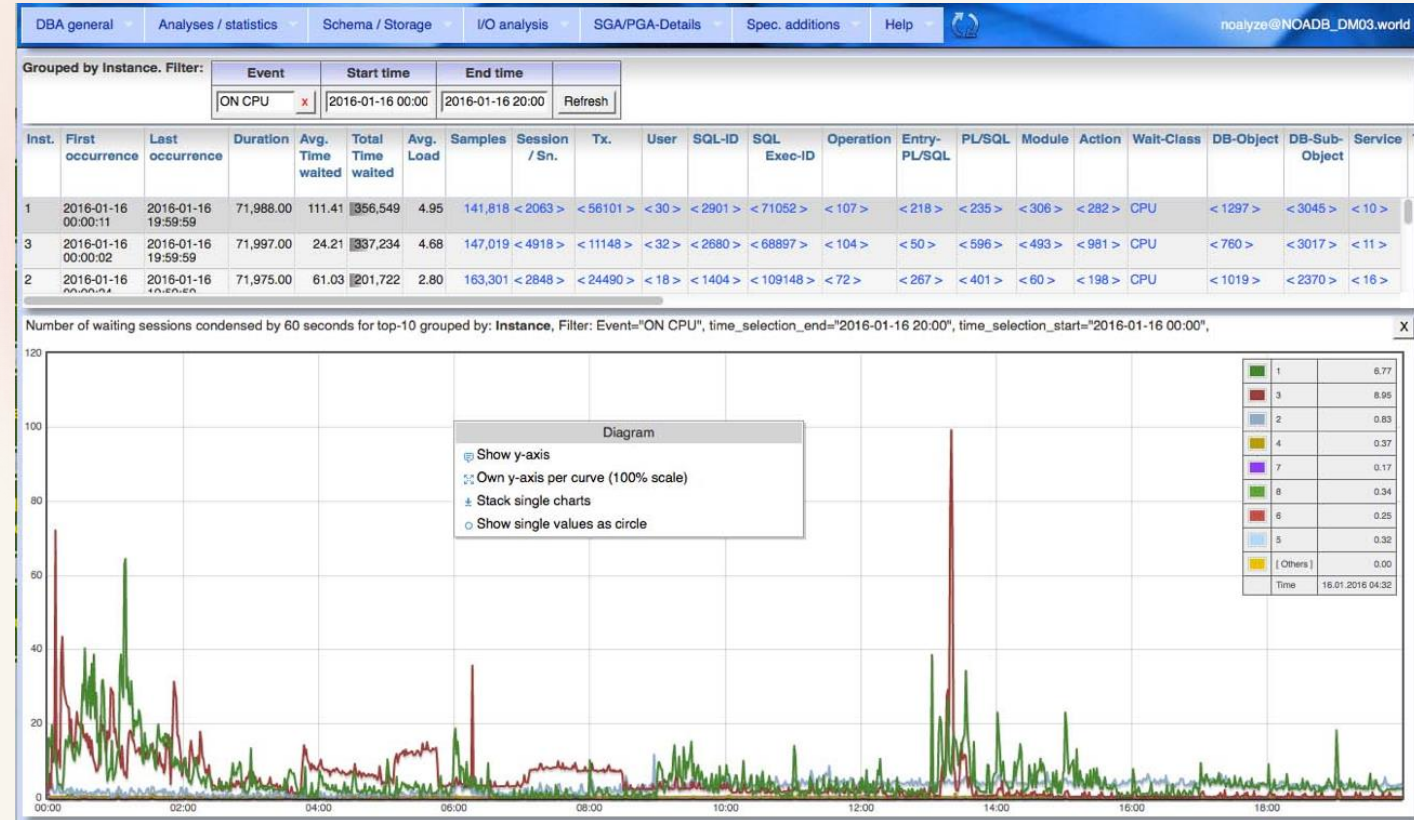
Open-Source Tool für Performance-Analyse von Oracle-DB, GPL3-Lizenz

Basiert auf per SQL erreichbaren Daten, Historien aus AWR-Daten oder eigenem Sampling

Nutzbar als Docker-Image oder selbststartendes war-File (Java)

Die GUI-Funktionen von Panorama greifen nur rein lesend auf die DB zu und installieren keine eigenen Objekte.

Beschreibung von Panorama inkl. Download-Link:
Blog zu Oracle-Performance-Analyse:
Diverse Slides zur Anwendung von Panorama:



<https://rammpeter.github.io/panorama.html>
<https://rammpeter.blogspot.com>
<https://speakerdeck.com/rammpeter>

SQL Plan Baselines

Vorgabe eines konkreten Ausführungsplanes per Plan Hash Value

- Bestandteil des SQL Plan Management für stabile Pläne
- Eigentlich ein optional mehrstufiges Konzept mit automatischer Erstellung und Freigabe über ENABLED, ACCEPTED, FIXED
- Hier heute reduziert auf eine fixed Baseline zum Pinnen eines bekannten Planes
- Der Optimizer verprobt die Hash Values der erstellten Pläne gegen evtl. Vorgaben in den Baselines und bevorzugt diese bei der Festlegung des endgültigen Planes
- Varianten zur Erzeugung einer SQL Plan Baseline:
 - Automatisches Capture bei erstmaligem Auftreten eines neuen Planes
 - Import aus anderer DB
 - Erzeugen aus aktuellem Cursor Cache der SGA per DBMS_SPM
 - Erzeugen aus AWR-Daten über SQL Tuning Set per DBMS_SQLSET

SQL Plan Baseline: Generieren per Panorama

Funktionen zum Generieren des Codes für Erzeugung von SQL Plan Baselines

- Aus aktuellen Ausführungsplänen in SGA per DBMS_SPM
 - Keine Management Pack Lizenz erforderlich
 - Auch für Standard Edition nutzbar (seit Rel. 18)
- Aus der AWR-Historie per DBMS_SQLSET
 - Erfordert Enterprise Edition + Diagnostics Pack (seit Rel. 18)
 - Vor Rel. 18 erfolgt Extraktion der Baseline per DBMS_SQLTUNE und erfordert Tuning Pack
- Diese Funktionen erlauben schnelle Rettung bei Problemen mit verändertem Ausführungsplan
 - Vorhandensein eines früheren besseren Ausführungsplanes reicht oft zur Lösung
 - Keine Notwendigkeit, das SQL inhaltlich zu verstehen
 - Aufwändigere Tuning-Maßnahmen können auf spätere Zeit verschoben werden
 - Plan Baselines in der Nutzungsart sollten jedoch keine Dauermaßnahme darstellen

SQL Plan Baseline: Generieren per Panorama: Step 1

SQL-Details anzeigen (aktuell in SGA oder Historie aus AWR)

Statement details of current SGA from GV\$SQL: Instance = 1, SQL-ID = 'avbqy2x751yz6', Child-No. = 1

```
SELECT * FROM doag.TestTable where id_mod_10000 = 496
```

Parsing schema name	SYSTEM	
Plan-Hash-Value / Opt.Env.	1538384467 / 3775051887	
Parsing module / action	SQL Developer	
First Load Time	2025-05-05 15:51:21	
Last Load Time	2025-05-05 16:01:23	
Last Active Time	2025-05-05 16:01:25	
Buffer cache hit ratio	100.00 %	
Metric-Name	Total	Per Exec.
Executions	4	-
Parse calls	4	1.00
Fetches	4	1.00
Sorts	0	0.00
Rows processed	200	50.00
Buffer gets	255	63.75
Disk reads	0	0.00
Elapsed Time (sec), incl. PQ Slaves	0	0.01
CPU-Time (sec)	0	0.01
Application Wait Time (sec)	0	0.00
Concurrency Wait Time (sec)	0	0.00
Cluster Wait Time (sec)	0	0.00
User-IO Wait Time (sec)	0	0.00
PL/SQL Exec Time (sec)	0	0.00

Execution-Plan (1)

Bind variables (0)

Objects (2)

Full history

Cursor Sharing (1)

Opt. Env.

Active Session History

Open Cursor (2)

Expand

DBMS_XPLAN

SQL-Monitor (0)

SQL plan mgmt.

SQL Plan Baseline: Generieren per Panorama: Step 2

Varianten des SQL Plan Management anzeigen, SQL Plan Baseline auswählen

Methods to generate external influence on the execution plan of SQL ID = 'avbqy2x751yz6' without changing the SQL syntax in application		
Action	Description	Option pack needed
Generate SQL Tuning Advisor Run SQL Tuning Advisor	Run Oracle's builtin SQL Tuning Advisor for this SQL to automatically find a better execution plan. ADVISOR privilege is needed for the user to run the SQL Tuning Advisor. CREATE ANY SQL PROFILE privilege is needed to create a SQL profile from the result of the Tuning Advisor.	Enterprise Edition + Diagnostics and Tuning Pack
SQL plan baseline	Generate script to fix exactly one execution plan as SQL-baseline for this SQL. Bases on current SGA data.	none
Generate script to fix exactly this execution plan as SQL-baseline for this SQL.		
SQL Profile	Generates commands for manual creation of a SQL profile. This allows to inject your own defined optimizer hints to the SQL.	Enterprise Edition + Tuning Pack
SQL Patch	Generates commands for creation of a SQL patch. This allows to inject your own defined optimizer hints to the SQL. Similar to SQL profile, but without the need for licensed Tuning Pack.	None
SQL Translation	Generates a script for using SQL translation framework. This allows to replace all parts of the SQL statement as long as the number and types of bind variables remain the same,	Enterprise Edition

SQL Plan Baseline: Generieren per Panorama: Step 3

Generierten Code ausführen als privilegierter User, optional Doku anpassen

```
-- Build SQL plan baseline for SQL-ID = 'avbqy2x751yz6', plan hash value = 1538384467
-- Generated with Panorama at 2025-05-05 18:32:59 +0200
-- Should be executed as user with DBA privileges

SET SERVEROUTPUT ON;

DECLARE
    cur          sys_refcursor;
    hit_count    PLS_INTEGER;
    sql_handle   VARCHAR2(30);
    plan_name    VARCHAR2(30);
BEGIN
    hit_count := DBMS_SPM.LOAD_PLANS_FROM_CURSOR_CACHE (
        sql_id      => 'avbqy2x751yz6',
        plan_hash_value => 1538384467,
        fixed       => 'YES',
        enabled     => 'YES'
    );

    -- Get access criteria for the new created baseline based on exact- or force-signature
    SELECT SQL_Handle, Plan_Name INTO sql_handle, plan_name FROM DBA_SQL_Plan_Baselines
    WHERE (Signature = 3311258233967618958 OR Signature = 13219411692035010115)
    AND Created > SYSDATE-0.1;

    -- Set the description according to your choice (max. 500 chars)
    hit_count := DBMS_SPM.ALTER_SQL_PLAN_BASELINE(SQL_Handle => sql_handle, Plan_Name => plan_name, Attribute_Name => 'description', Attribute_Value => 'Baseline generated by Panorama');

    -- You can check the existence of the baseline now by executing:
    -- SELECT * FROM dba_sql_plan_baselines;
    -- or by looking at SQL details page for your SQL with Panorama

    -- Next commands remove currently existing cursors of this SQL from SGA to ensure hard parse with SQL plan baseline at next execution

    FOR cu IN (SELECT RAWTOHEX(Address) Address, Hash_Value FROM v$SQLArea WHERE SQL_ID='avbqy2x751yz6') LOOP
        DBMS_SHARED_POOL.PURGE (cu.address||', '||cu.hash_value, 'C');
        DBMS_OUTPUT.PUT_LINE('Existing cursor for SQL-ID=avbqy2x751yz6 removed from SGA');
    END LOOP;
END;
```

Beeinflussen des Optimizers bei der Erzeugung von Ausführungsplänen

- Externes Zusteuern von Optimizer Hints und Statistiken zu SQL
- Ein SQL Profile besteht aus einer Menge von konkreten Optimizer Hints (Outlines)
- Nutzung erfordert Enterprise Edition und Lizenzierung des Tuning Packs
- Varianten zur Erzeugung eines SQL Profile:
 - Ergebnis der Ausführung des SQL Tuning Advisors
 - Import per DBMS_SQLTUNE

SQL Profile : Generieren per Panorama

Funktionen zur Erzeugung von SQL Profiles

1. Generieren des Codes für Ausführung des SQL Tuning Advisors mit Ausgabe der Empfehlungen
2. Direkte Ausführung des SQL Tuning Advisors in Panorama mit optionaler Erzeugung eines SQL Profiles mit den empfohlenen Hints
3. Generieren des Codes für manuellen Import konkreter Hints per `DBMS_SQLTUNE.Import_SQL_Profile`

SQL Profile : Generieren des SQL Tuning Advisor Codes für SQL

PL/SQL Exec Time (sec) 14,959 0:00

Execution-Plan (1) Bind variables (2) Objects (1) Full history Cursor Sharing (1) Opt. Env. Active Session History Open Cursor (63) Expand DBMS_XPLAN SQL-Monitor (0) SQL plan mgmt.

Methods to generate external influence on the execution plan of SQL ID = 'crbw071kcpzwa' without changing the SQL syntax in application

Action	Description	Option pack needed
Generate SQL Tuning Advisor	Run Oracle's builtin SQL Tuning Advisor for this SQL to automatically find a better execution plan.	Enterprise Edition + Diagnostics and Tuning Pack
Run SQL Tuning Advisor		
SQL plan baseline	Generate script to fix exactly one execution plan as SQL-baseline for this SQL. Bases on current SGA data.	none

```
-- Run SQL Tuning Advisor SQL-ID = 'crbw071kcpzwa'
-- Generated with Panorama at 2025-05-06 12:18:20 +0200

SET SERVEROUTPUT ON;

DECLARE
    Task_Name VARCHAR2(200);
BEGIN
    Task_Name := DBMS_SQLTUNE.create_tuning_task(
        sql_id      => 'crbw071kcpzwa',
        scope       => 'COMPREHENSIVE',
        time_limit  => 60, /* adjust to your needs */
        task_name   => 'Panorama_crbw071kcpzwa',
        description => '<my description of task>'
    );

    DBMS_SQLTUNE.execute_tuning_task(Task_Name);
    DBMS_OUTPUT.PUT_LINE(DBMS_SQLTUNE.report_tuning_task(Task_Name, /* Type*/ 'TEXT', /* Level */ 'ALL', owner_name => USER ));
END;
/

-- to accept a SQL profile provided by this SQL Advisor tuning task execute:
-- EXEC DBMS_SQLTUNE.accept_sql_profile(task_name => 'Panorama_crbw071kcpzwa', name => 'Panorama_crbw071kcpzwa', task_owner => USER, replace => TRUE);

-- to drop the SQL Advisor tuning task as last step execute:
-- EXEC DBMS_SQLTUNE.drop_tuning_task('Panorama_crbw071kcpzwa');
```

SQL Profile : Ausführen des SQL Tuning Advisors für SQL

Run Advisor, check suggestions, accept SQL Profile or not, drop tuning task

Execution-Plan (1) Child summary (2) Bind variables (0) Objects (1) Full history Cursor Sharing (1) Opt. Env. Active Session History Open Cursor (2) Expand DBMS_XPLAN SQL-Monitor (0) SQL plan mgmt.

Methods to generate external influence on the execution plan of SQL ID = 'avbqy2x751yz6' without changing the SQL syntax in application

Action	Description	Option pack needed
Generate SQL Tuning Advisor Run SQL Tuning Advisor	Run Oracle's builtin SQL Tuning Advisor for this SQL to automatically find a better execution plan. ADVISOR privilege is needed for the user to run the SQL Tuning Advisor. CREATE ANY SQL PROFILE privilege is needed to create a SQL profile from the result of the Tuning Advisor.	Enterprise Edition + Diagnostics and Tuning Pack
SQL plan baseline	Generate script to fix exactly one execution plan as SQL-baseline for this SQL. Bases on current SGA data.	none
SQL Profile	Generates commands for manual creation of a SQL profile. This allows to inject your own defined optimizer hints to the SQL.	Enterprise Edition + Tuning Pack

Run SQL Tuning Advisor for SQL ID = 'avbqy2x751yz6'

Name Description Time limit Overwrite existing? ☐ Scope [Execute SQL Tuning Advisor](#)

Result of SQL Tuning Advisor task execution for task name = 'Panorama_avbqy2x751yz6', SQL ID = 'avbqy2x751yz6'

[Accept recommended SQL Profile](#) [Drop SQL Tuning Advisor task](#)

GENERAL INFORMATION SECTION

Tuning Task Name : Panorama_avbqy2x751yz6
Tuning Task Owner : DOAG
Tuning Task ID : 2499
Workload Type : Single SQL Statement
Execution Count : 1
Current Execution : EXEC_2537
Execution Type : TUNE SQL
Scope : COMPREHENSIVE
Time Limit(seconds): 60
Completion Status : COMPLETED
Started at : 05/06/2025 12:07:28
Completed at : 05/06/2025 12:07:29

Schema Name : SYSTEM

SQL Profile :Generieren Template Code für manuelle Definition

Execution-Plan (1) Bind variables (0) Objects (2) Full history Cursor Sharing (1) Opt. Env. Active Session History Open Cursor (2) Expand DBMS_XPLAN SQL-Monitor (0) SQL plan mgmt.

Methods to generate external influence on the execution plan of SQL ID = 'avbqy2x751yz6' without changing the SQL syntax in application

Action	Description	Option pack needed
	Bases on current SGA data.	
SQL Profile	Generates commands for manual creation of a SQL profile. This allows to inject your own defined optimizer hints to the SQL.	Enterprise Edition + Tuning Pack
SQL Patch	Generates commands for creation of a SQL patch. This allows to inject your own defined optimizer hints to the SQL. Similar to SQL profile, but without the need for licensed Tuning Pack.	None

```
-- Script for establishing SQL profile for SQL-ID='avbqy2x751yz6'
-- Generated by Panorama at 2025-05-06 14:36:05 +0200
-- Executing this script allows you to add optimizer hints that are used for execution of this SQL.
-- This allows you to transparently influence execution plan without any change of the calling application.
-- Existing SQL-profiles for SQLs are shown by Panorama in SQL-details view.
-- All existing SQL-profiles are listed in Panorama via menu 'SGA/PGA-details' / 'SQL plan management' / 'SQL profiles'

-- Attributes that must be adjusted by you in this script for parameters:
--   - 'hint_text'   Place your optimizer hints here.
--                   Be aware that query block names and table-aliases must be used in optimizer hints
--                   with same values as they appear in columns QBLOCK_NAME and OBJECT_ALIAS of v$SQL_PLAN!
--                   Panorama shows query block name and alias in execution plan view by mouse over hint on column "Object name".
--                   Example: "FULL(@SEL$E029B2FF tab@SEL$2)" where "tab" ist the table alias used in SQL-statement
--   - 'decription' describe purpose of SQL profile

-- ##### To establish SQL profile execute this as SYSDBA #####
-- on Pluggable database execute it connected to PDB, not CDB

-- Drop already existing SQL profile for this SQL before applying new patch
EXEC DBMS_SQLTUNE.Drop_SQL_Profile('Panorama_avbqy2x751yz6');

BEGIN
  DBMS_SQLTUNE.Import_SQL_Profile(
    sql_text  => 'SELECT * FROM doag.TestTable where id_mod_10000 = 496',
    name      => 'Panorama-Profile avbqy2x751yz6',
    profile   => SQLPROF_ATTR('< my personal hint>'),
    description => 'My personal description for profile',
    force_match => TRUE
  );
END;
/

-- ##### To remove the SQL-profile if not needed anymore execute this as SYSDBA #####
-- EXEC DBMS_SQLTUNE.Drop_SQL_Profile('Panorama_avbqy2x751yz6');
```


Beeinflussen des Optimizers bei der Erzeugung von Ausführungsplänen

- Wirkungsweise ist analog zu SQL Profile über Optimizer Hints
- SQL Patch erfordert jedoch keine Managment Pack Lizenz
- Auch anwendbar in Standard Edition
- Varianten zur Erzeugung eines SQL Patch:
 - In Rel. 11g per `DBMS_SQLDIAG_INTERNAL.I_CREATE_PATCH`
 - Ab Rel. 12.1 per SQL Repair Advisor (Package `DBMS_SQLDiag`)
 - Manueller Import per `DBMS_SQLDiag.create_SQL_patch`

SQL Patch :Generieren Template Code für manuelle Definition

Execution-Plan (1) Bind variables (0) Objects (1) Full history Cursor Sharing (1) Opt. Env. Active Session History Open Cursor (3) Expand DBMS_XPLAN SQL-Monitor (0) SQL plan mgmt.

Methods to generate external influence on the execution plan of SQL ID = 'avbqy2x751yz6' without changing the SQL syntax in application

Action	Description	Option pack needed
SQL Patch	Generates commands for creation of a SQL patch. This allows to inject your own defined optimizer hints to the SQL. Similar to SQL profile, but without the need for licensed Tuning Pack.	None
SQL Translation	Generates a script for using SQL translation framework. This allows to replace all parts of the SQL statement as long as the number and types of bind variables remain the same,	Enterprise Edition

```
-- Script for establishing SQL patch for SQL-ID='avbqy2x751yz6'
-- Generated by Panorama at 2025-05-06 16:45:16 +0200
-- Executing this script allows you to add optimizer hints that are used for execution of this SQL.
-- This allows you to transparently influence execution plan without any change of the calling application.
-- Existing SQL-patches for SQLs are shown by Panorama in SQL-details view.
-- All existing SQL-patches are listed in Panorama via menu 'SGA/PGA-details' / 'SQL plan management' / 'SQL patches'

-- Attributes that must be adjusted by you in this script for parameters:
--   - 'hint_text'   Place your optimizer hints here.
--                   Be aware that query block names and table-aliases must be used in optimizer hints
--                   with same values as they appear in columns QBLOCK_NAME and OBJECT_ALIAS of v$SQL_PLAN!
--                   Panorama shows query block name and alias in execution plan view by mouse over hint on column "Object name"
--                   or as additional columns "Query block name" and "Object alias".
--                   Example: "FULL(@SEL$E029B2FF tab@SEL$2)" where "tab" ist the table alias used in SQL-statement
--   - 'decription' describe purpose of SQL-patch

-- ##### To establish SQL patch execute this as SYSDBA #####
-- on Pluggable database execute it connected to PDB, not CDB

DECLARE
  patch_name VARCHAR2(32767);
BEGIN
  patch_name := sys.DBMS_SQLDdiag.create_SQL_patch(
    sql_id    => 'avbqy2x751yz6',
    hint_text => '< my personal hint>',
    name      => 'Panorama-Patch avbqy2x751yz6',
    description => 'My personal description for patch'
  );
END;
/

-- ##### To remove the SQL-patch if not needed anymore execute this as SYSDBA #####
-- EXEC DBMS_SQLDdiag.Drop_SQL_Patch('Panorama-Patch avbqy2x751yz6');
```

SQL Translation

Austauschen der Syntax von SQL-Statements vor dem Parsen

- Komplettes Ersetzen von konkreten SQL-Statements durch Alternative
- Anforderungen an Ersatz-SQL für Kompatibilität mit Applikation:
 - Gleiche Anzahl, Namen, Reihenfolge und Typen der Spalten im Result
 - Gleiche Anzahl, Reihenfolge und Typen der Bindevariablen
- Erfordert Enterprise Edition
- Extremes Potential zur nachträglichen Korrektur der kompletten SQL-Syntax
 - Z.B. Austausch von Tabellennamen, Einfügen weitere Joins etc.
- Hohes Risiko von Konfusion bei der Analyse der ausgeführten SQLs
- Schritte zur Nutzung von SQLTranslations:
 - Erzeugen eines SQL Translation Profile erlauben für den ausführenden User
 - Erzeugen des SQL Translation Profile durch den diesen User
 - Erzeugen eines LOGON-Triggers für diesen User zur Profile-Aktivierung

SQL Translation: Generiertes Code-Template für Erzeugung

```
-- ##### Following activities should be sequentially executed in this order to establish translation #####

-- 1. ##### Execute as DBA to establish translation:

GRANT CREATE SQL TRANSLATION PROFILE TO SYSTEM;

GRANT ALTER SESSION TO SYSTEM;

-- 2. ##### Execute as user SYSTEM to establish translation:

EXEC DBMS_SQL_TRANSLATOR.CREATE_PROFILE('PANORAMA_TRANSL_SYSTEM');

BEGIN
DBMS_SQL_TRANSLATOR.REGISTER_SQL_TRANSLATION('PANORAMA_TRANSL_SYSTEM',
'SELECT * FROM doag.TestTable where id_mod_10000 = 496',
-- ##### Adjust the following SQL text as translation target on your needs
'SELECT * FROM doag.TestTable where id_mod_10000 = 496',
TRUE);
END;
/

-- 3. ##### Execute as SYS as SYSDBA to establish translation:
-- If not created as SYS you'll get "ORA-01031: insufficient privileges" at execution of "ALTER SESSION SET SQL_TRANSLATION_PROFILE"
-- Check alert log for errors from logon trigger if translation does not work in new session

CREATE TRIGGER Panorama_Transl_SYSTEM AFTER LOGON ON DATABASE
BEGIN
-- created by SQL translation script generated by Panorama to allow translation of SQL with SQL-ID='avbqy2x751yz6'
IF USER = 'SYSTEM' THEN
EXECUTE IMMEDIATE 'ALTER SESSION SET CURRENT_SCHEMA=SYSTEM';
EXECUTE IMMEDIATE 'ALTER SESSION SET SQL_TRANSLATION_PROFILE=PANORAMA_TRANSL_SYSTEM';
EXECUTE IMMEDIATE 'ALTER SESSION SET EVENTS =''10601 trace name context forever, level 32''';
END IF;
END;
/
```


Übersicht über existierende Vorgaben im SQL Plan Management

DBA general ▾

Analyses / statistics ▾

Schema / Storage ▾

I/O analysis ▾

SGA/PGA-Details ▾

Spec. additions ▾

Help ▾

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SQL plan management directives

Config attributes SQL profiles (0) SQL plan baselines (7) Stored outlines (0) SQL patches (0) SQL Tuning Advisor tasks (0)

SQL plan baselines from

SQL handle / plan name	Creator	Origin	Parsing schema name	Description	Version	Last executed	Last verified	Enbl.	Acc.	Fixed	Repr.	Auto purge	Optim. costs	Usage
SQL_b77123aa0f903e6f SQLID_9vcdtrp717psy_4040321520	INWMS	MANUAL-LOAD-FROM-CURSOR-CACHE	INWMS		19.0.0.0.0	2025-04-24 09:02:10		YES	YES	YES	YES	YES	3,128	
SQL_026b85511c0f7592 SQL_PLAN_04uw5a4f0yxck297c4da7	INWMS	MANUAL-LOAD-FROM-CURSOR-CACHE	INWMS		19.0.0.0.0	2025-04-24 06:15:31		YES	YES	YES	YES	YES	11,276	4
SQL_9235adcdfa377589 SQL_PLAN_94dddtrx3fxc9ad8b07f5	INWMS	MANUAL-LOAD-FROM-CURSOR-CACHE	INWMS		19.0.0.0.0			YES	YES	NO	YES	YES	2	c
SQL_9235adcdfa377589 SQL_PLAN_94dddtrx3fxc95394c2d3	INWMS	MANUAL-LOAD-FROM-CURSOR-CACHE	INWMS		19.0.0.0.0	2024-05-13 15:33:17	2024-05-13 15:35:07	NO	YES	NO	YES	YES	2	
SQL_026b85511c0f7592 SQLID_crts9gyk3pdub_2922348656	INWMS	MANUAL-LOAD-FROM-CURSOR-CACHE	INWMS		19.0.0.0.0	2025-04-08 11:54:52	2025-04-08 11:54:54	YES	YES	YES	NO	YES	12,727	

SQL-Area

SQL-Area day comparison

SGA Memory

DB-Cache

Object usage by SQL

PGA-statistics

Result Cache

SQL plan management

Show all SQL plan management directives of database:

- SQL profiles
- SQL plan baselines
- Stored outlines
- SQL translations
- SQL patches

Fazit

- **Schnelle Reaktion auf problematische SQL ist möglich auch ohne Änderung der Syntax**
- **Quick fixes auch als solche Betrachten**
- **Helfen die Nacht zu überstehen, sind aber meist keine Dauerlösung**
- **Ausnahmen:**
 - **Plan Baselines als Strategie für fixierte Pläne**
 - **Stabile SQL-Syntax garantiert**

Vielen Dank für Ihr Interesse



Peter Ramm

Team Lead Architecture Consulting

[Contact me via Teams](#)

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Weitere interessante Quellen:

- [Osman DiNÇ: SQL Plan Baseline, SQL Patch, SQL Profile: Differences and Use Cases](#)